

Laboratory work 1

Linear algorithms

- ✓ **Introduction to the programming environment MS Visual Studio**
- ✓ **Program structure in C language**
- ✓ **Types of C language data**
- ✓ **Variables and expressions**
- ✓ **I/O consol**

Objective : To study the structure of the C language program; learn how to work with the MS Visual Studio family programming environment; to get acquainted with I/O operators; learn how to use simple data types in a program, compile simple block diagrams of programs and programs in C.

Short theoretical information

1.1. Familiarity with the programming environment MS Visual Studio

The program, which is created in the MS Visual Studio environment, is organized as a project. A project is a collection of interconnected output files designed to solve a particular task. The project includes files created by the programmer, as well as files created automatically by the programming environment. Compiling and composing project files allows you to get an executable program. In conjunction with the creation of the project, MS Visual Studio creates a solution as a means of combining multiple projects to make them easier to work with.

The MS Visual Studio programming environment supports multiple programming languages and software platforms, so you must complete the following preparatory steps before you begin writing a C language program:

- Launch MS Visual Studio 2017. Upon its launch, the start window shown in Fig. 1.1. Select **File** → **New** → **Project** main menu item or press **Ctrl + Shift + N** as shown in the figure. 1.2;
- In the **New Project** window, expand **Visual C ++** node, select **Win32** item in the center panel, select the type of **Win32 Console Application** project created, as shown in fig. 1.3;
- Enter a project name in the **Name** field, such as **lab1**;
- In the **Location** field, enter the name of the directory that will host the project and the full path to it, such as **D:\student**. The directory can also be selected in the **Project Location** dialog box, which can be opened using the **Browse...** button;
- Press the **OK** button;

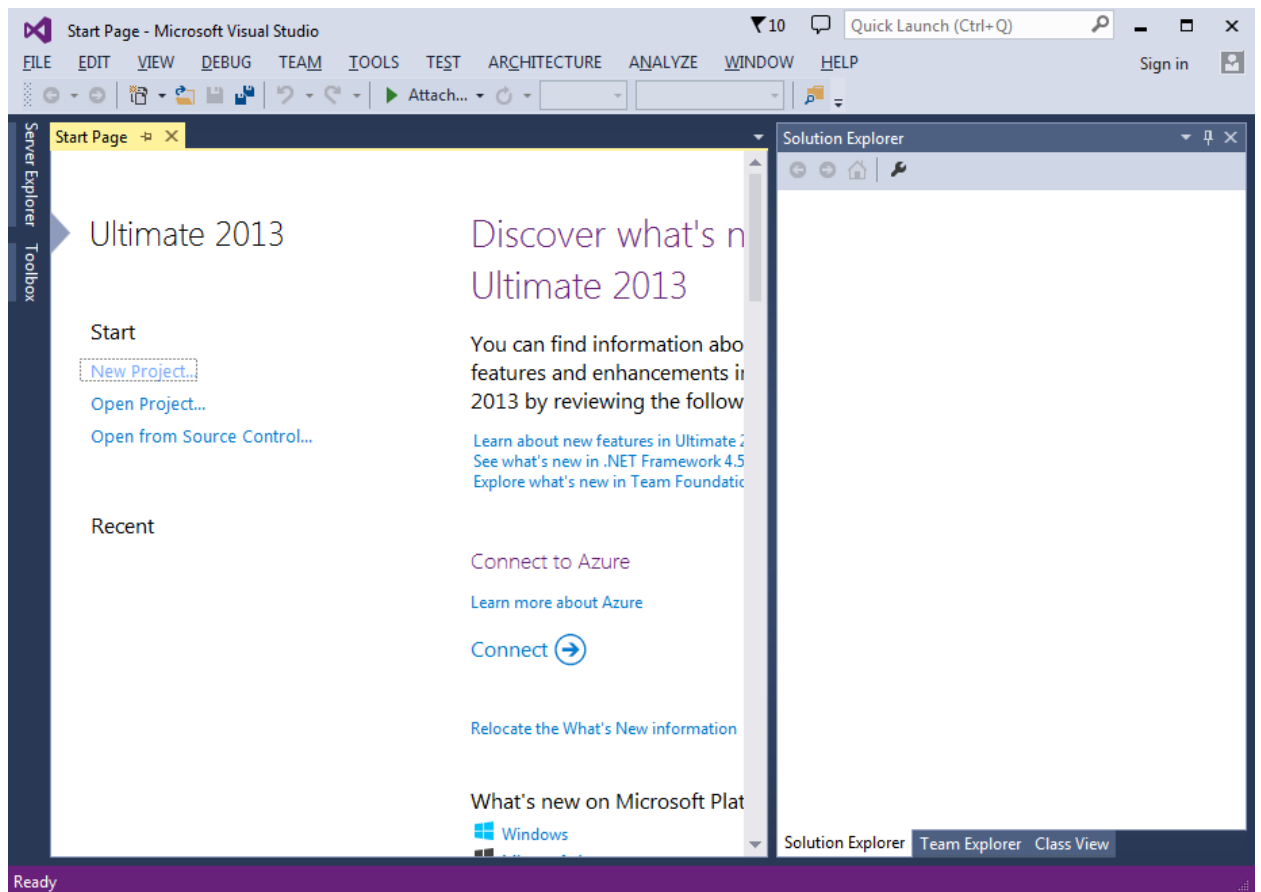


Fig. 1.1 - Visual Studio 2017 launcher

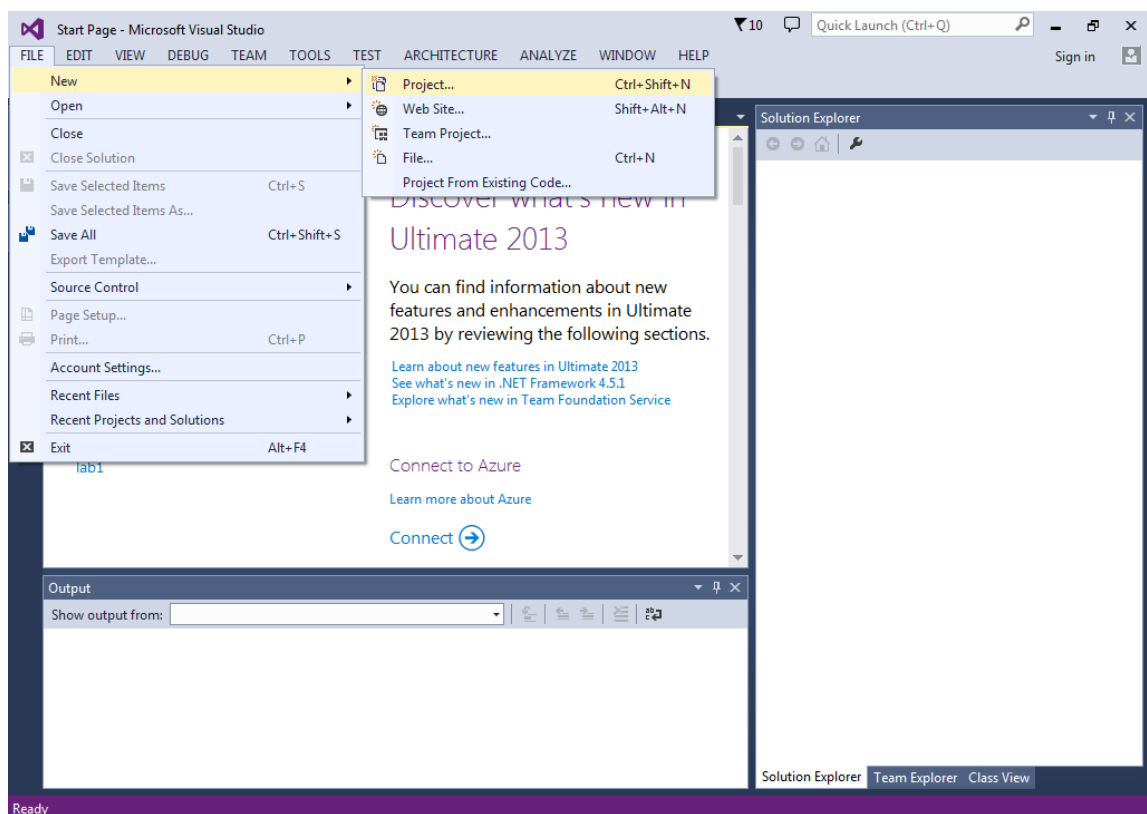


Fig. 1.2 - Create a new project in Visual Studio 2017

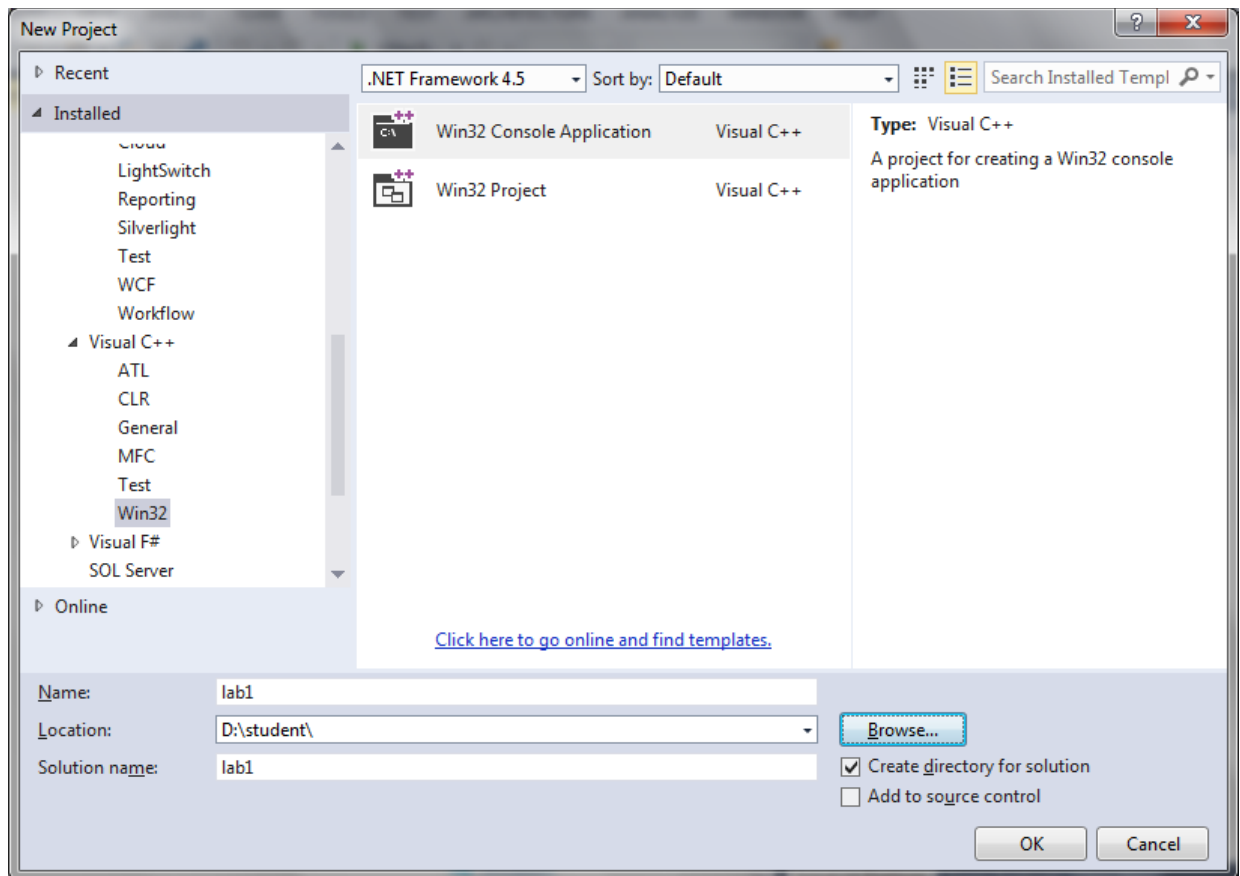


Fig. 1.3 - Selection of the project type

- In the window of the **Win32 Application Wizard** (after clicking the **Next** button) in additional options, select the **Empty project** option, as shown in fig. 1.4 and click **Finish**.

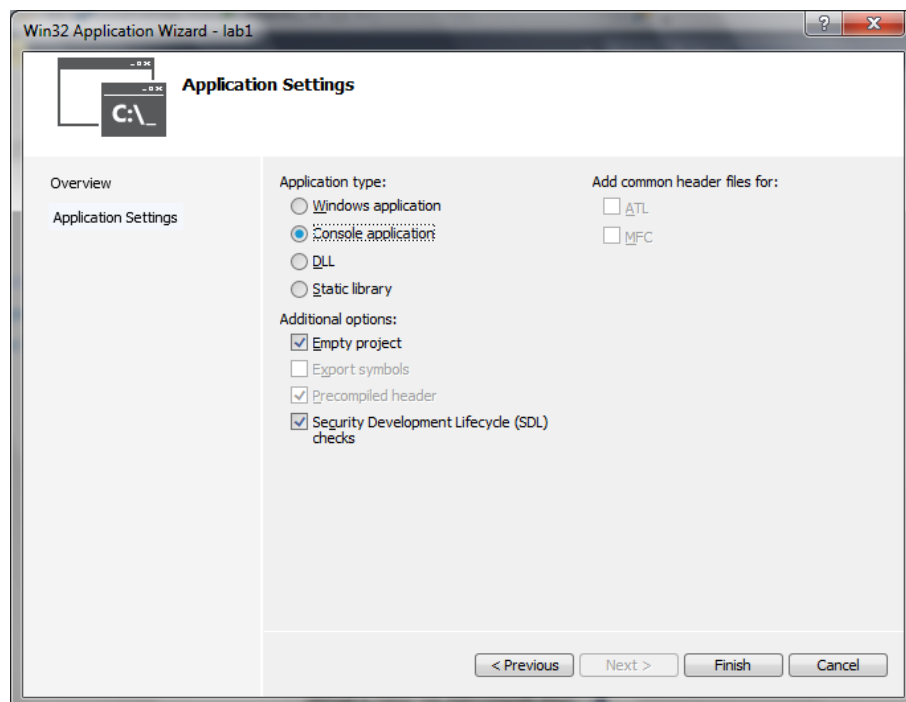


Fig. 1.4 - Advanced customization options

To add a source code file to the project, perform the following steps in the **Solution Explorer** window (Figure 1.5):

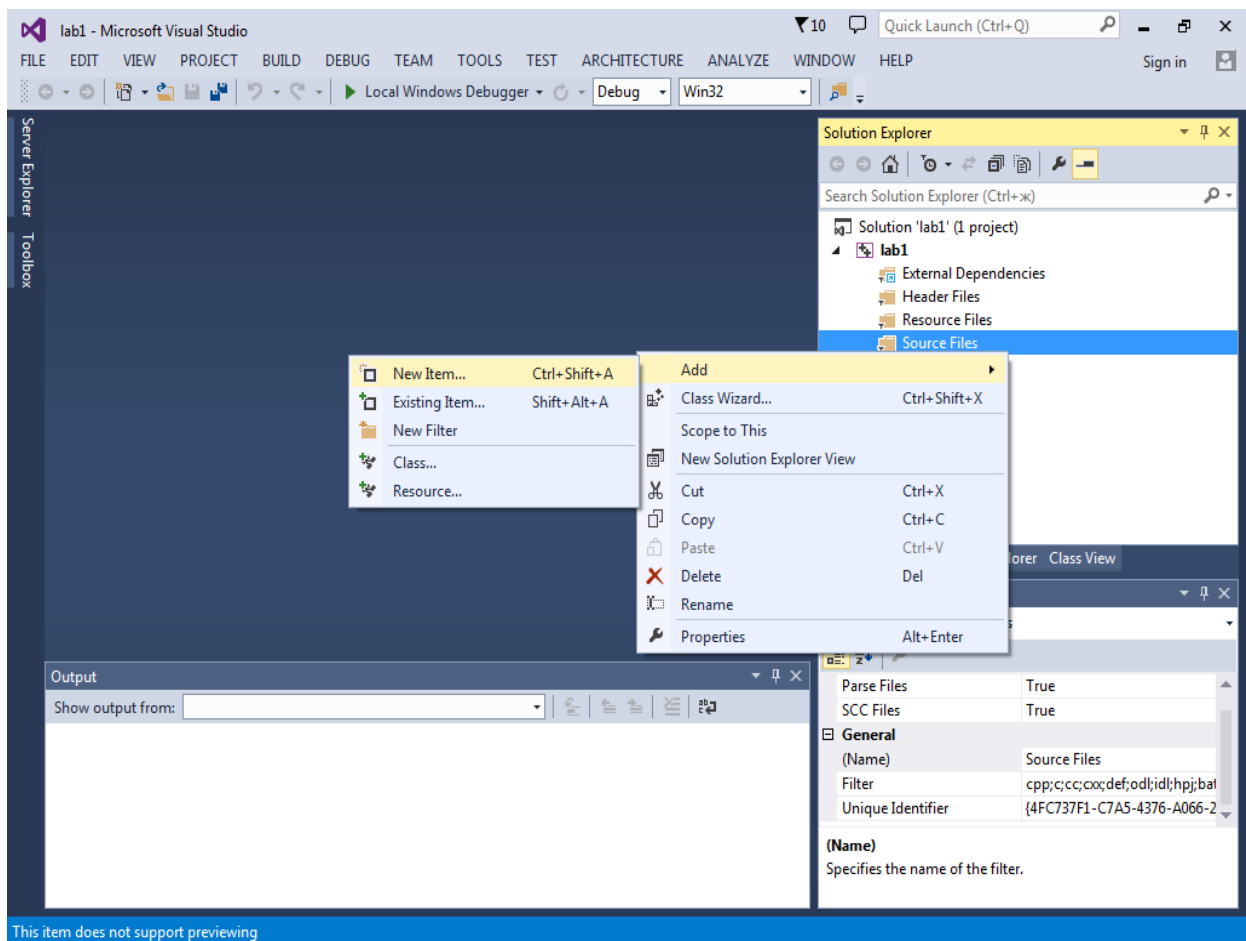


Fig. 1.5 - Adding to the project file

- Right-click on the **Source Files** folder (node lab1) and select **Add → New Item** from the pop-up menu;
- In the **Add New Item** window, select the **Visual C ++ Node Code** menu item, in the center of the panel select the **C ++ File (.cpp)** file type. In the **Name** field, specify a file name, such as **main.c**, where the **.c** extension indicates that the file contains code in C (Fig. 1.6);
- Click the **Add** button.

The added file will be displayed in the **Solution Explorer** tree (**Source Files** node, source files). For the created file, the code editor is automatically opened, the window of which is shown in fig. 1.7.

To set up a project, follow these steps:

- Select the menu item **Project → lab1 Properties** (Fig. 1.8);
- In the list of **C/C++** node, select **Advanced**, in the right part of the panel, set **Compile As** property to **Compile as C Code (/ TC)**, as shown in fig. 1.9;
- Click the **Apply** and **OK** buttons.

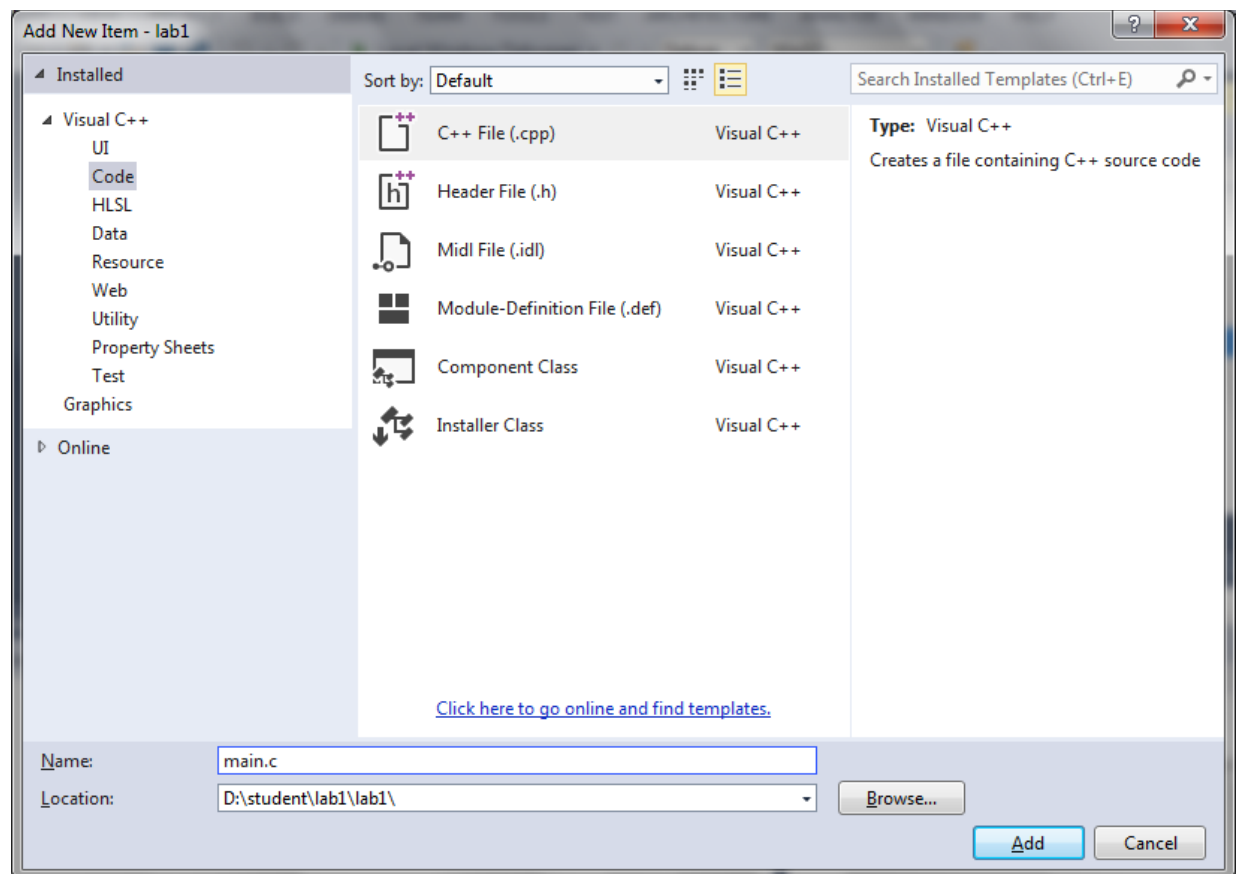


Fig. 1.6 - Main.c extension

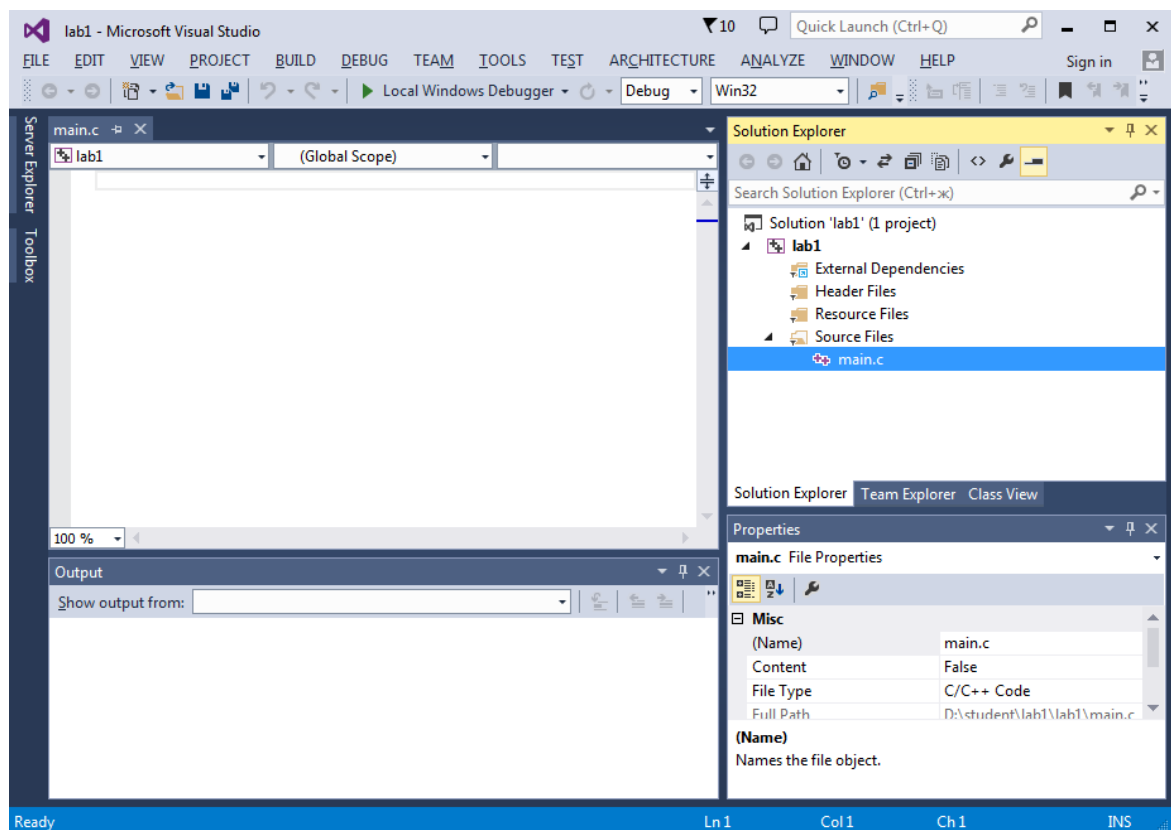


Fig. 1.7 - Create a project file

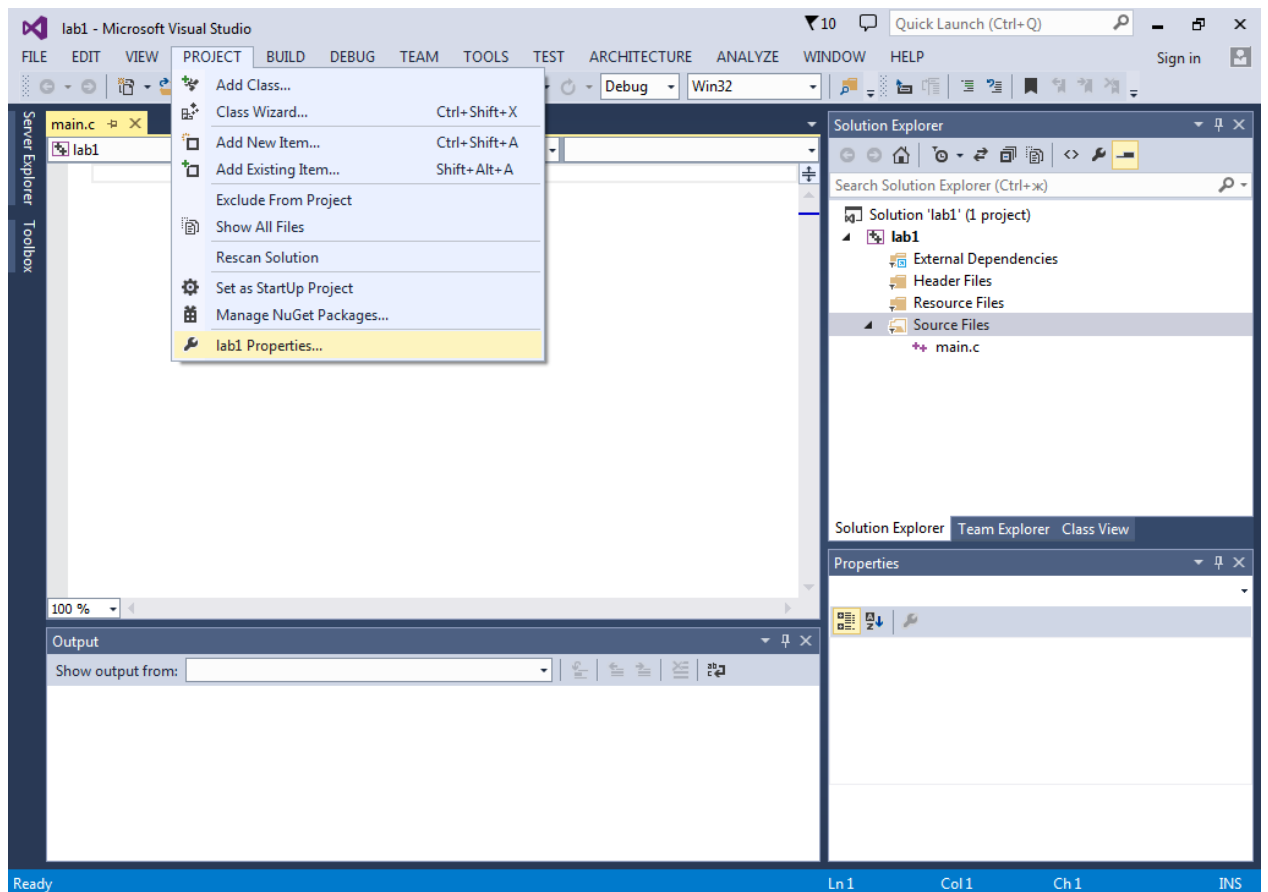


Fig. 1.8 - Main menu item

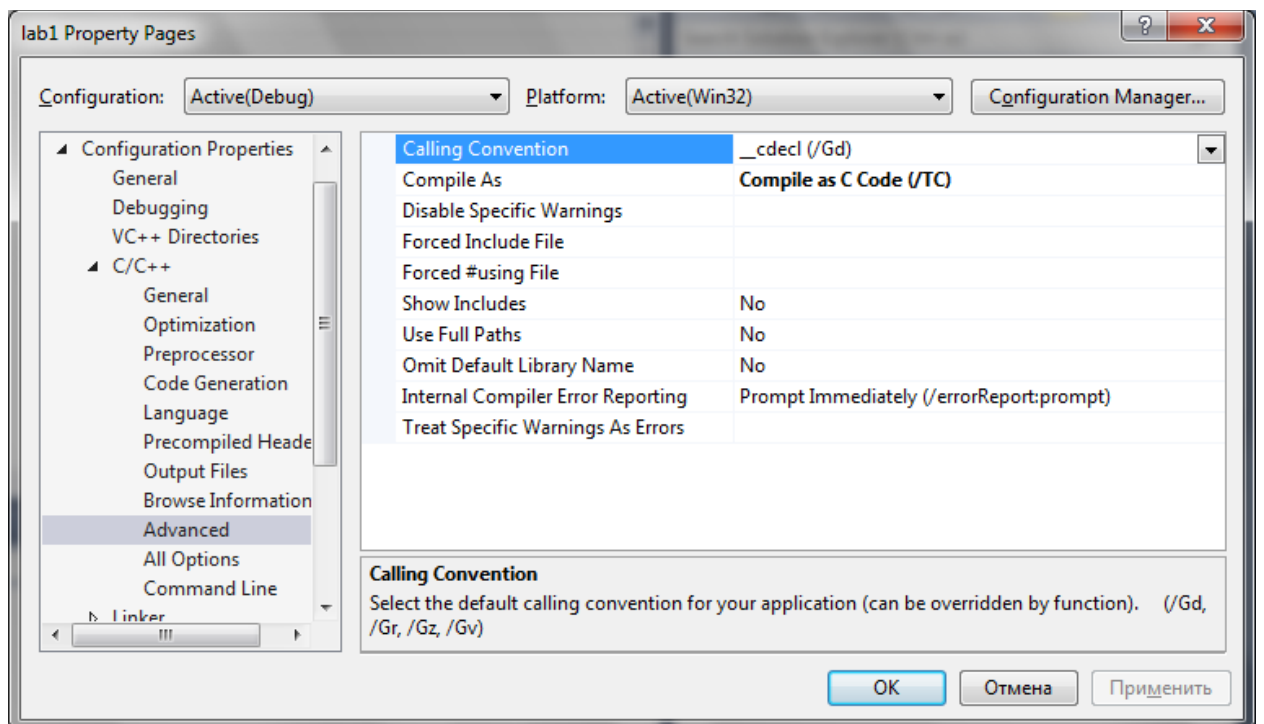


Fig. 1.9 - Installation of Compile as C Code (/TC)

The **Solution Explorer** window displays folders that contain project files:

- **External Dependencies** contains files that were not added to the project, but used in source files (for example, added using the **#include** directive);
- **Header Files** contains **.h** header files;
- **Resource Files** contains resource files;
- **Source Files** contains files with source code (in this project, files with **.c** extension).

To create an executable file, select the **Build → Build Solution** menu item. To start the program, select the **Debug → Start Without Debugging** or press **Ctrl + F5**.

Tasks of laboratory work

Task 1

On the given sides a , b , c of a triangle (fig. 1.10) to find sizes according to variants. Provide input values a , b , c from the keyboard. Output of initial data to carry out with the corresponding explanatory text. Develop an algorithm and program for finding the values specified in table. 1.1.

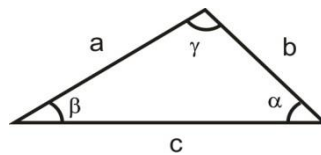


Fig. 1.10 – Triangle

Table 1.1 – Individual tasks

Individual tasks	Calculated value	Initial data		
		a	b	c
1	Angle α and height h	15.87	12.9	18.56
2	Angle β and median m	92.0	53.29	90.1
3	Angle γ and bisector w	32.0	28.75	39.11
4	Angle α and radius R	34.5	17.32	34.71
5	Angle β and median m	13.2	10.8	8.37
6	Angle γ and height h	44.73	50.21	65.98
7	Angle α and radius r	51.56	31.76	54.87
8	Angle β and height h	19.65	25.87	29.53
9	Angle γ and bisector w	38.93	42.21	34.42
10	Angle α and bisector w	60.1	88.47	95.12

Theoretical information

1. The formula for the area of a triangle

$$S = 1/2 \cdot a \cdot h$$

2. Heron's formula

$$S = \sqrt{p(p-a)(p-b)(p-c)},$$

where p is half of the perimeter $p = (a + b + c)/2$.

3. Basic trigonometric relations

$$\begin{aligned} \arcsin x &= \operatorname{arctg}(x / \sqrt{1-x^2}); & \alpha &= \arccos((b^2 + c^2 - a^2) / (2bc)); \\ \arccos x &= \operatorname{arctg}(\sqrt{1-x^2} / x); & \beta &= \arccos((a^2 + c^2 - b^2) / (2ac)); \\ \operatorname{arctg} x &= \operatorname{arctg}(1 / x); & \gamma &= \arccos((a^2 + b^2 - c^2) / (2ab)). \end{aligned}$$

Task 2

Develop the calculation algorithm and the C language program to determine the values of the function:

$$F = \frac{E1^2 + |E2| + E3}{\sqrt{|E1 \cdot E3|}} - \frac{1}{3}.$$

The functions $E1$, $E2$, $E3$ according to the options are given in table 1.2. Output of source data should be carried out with the appropriate explanatory text and formatted output.

Table 1.2 – Individual tasks

Individual tasks	$E1$	$E2$	$E3$
1	$\ln (x+3)/a $	$\operatorname{tg}(ax/b)$	$(a^{-x} + b^x)/3$
2	$(e^{ax} - e^{bx})/2b$	$(e^{ax} + e^{bx})/2a$	$(e^{abx} + e^{ax/b})/2$
3	$\ln(\operatorname{acos}^2 x)/(x^3 + 1)$	$(bx^2 + a) / (ax^2 + bx - 3)$	$(\sin x \cdot \cos ax)/(ax + b)$
4	$(ae^{\cos x})/b$	$x + 2abx + 4$	$(a + b) (\cos x /ab)^{-2}$
5	$\sin(\ln x-l)$	$\ln \sin x - 1 $	$(a + b) (ax + b /2)^{-0.5}$
6	$ae^{-\sin x}$	$b(x-1)/(x^2+1)$	$ a+b ^{x-a}$
7	$\sin(x+e^{\sin x})$	$e^{-x+b} + e^{-x+a}$	$(b-ax^3)/(a-bx^3)$
8	$x^3 + ax^2 + bx + 1$	$\operatorname{tg}(x + \cos x)$	$\ln x+ab /(x-ab)$
9	$\sqrt[3]{ ax + b - 3 }$	$\sqrt[3]{ bx - b + 4 }$	$e^{-bx/(a+3)}$
10	$a + e^{\sin(x-a)}$	$b - e^{\sin(x-b)}$	$\operatorname{tg}(e^{\sin x} + e^{\cos x})$

Task 3

By using Horner's algorithm, compile a program to calculate the value of the polynomial. According to the compiled program to calculate the value of the polynomial at a given value X. Output of the original data to carry out with the appropriate explanatory text and formatted output. Individual tasks are given in table 1.3.

Table 1.2 – Individual tasks

Individual tasks	Polynomial	x
1	$y = 3.2(1 + x)^6 + 5.1(1 + x)^5 - (1 + x)^2 - 7$	-4.3
2	$z = 6.3\cos^5 x - 4.2\cos^4 x + 8.1\cos^2 x - 1.5\cos x$	2.7
3	$z = -3.2\sqrt{(1+x)^7} - 1.1\sqrt{(1+x)^2} + 7.5\sqrt{(1+x)^2} + 4.3$	0.5
4	$p = -0.5\sin^7 2x + 3.3\sin^4 2x - 1.2\sin 2x - 5.4$	-1.3
5	$u = 8.1x^2 - 3.4x\sqrt{x} + 4.5\sqrt{x} + 1.5$	10.8
6	$t = 4.3tg^5 x - 6.1tg^4 x + 2.7tg^3 x - tgx + 3.2$	-0.8
7	$p = tg^5(1 + x) - 4.7tg^4(1 + x) + 6.3tg(1 + x) - 3.8$	19.7
8	$s = 2^{7x} - 8.3 \cdot 2^{5x} + 4.1 \cdot 2^{3x} - 6.3 \cdot 2^{2x} + 1.5 \cdot 2^x$	1.57
9	$c = -3.7(1 + \sqrt{x})^6 + 4.1(1 + \sqrt{x})^3 - 2.1(1 + \sqrt{x}) + 1.2$	30.5
10	$r = 3.2\ln^5 x - 4.5\ln^4 x + \ln^2 x - 7.3\ln x$	0.75

Task 4

Find the area of the shaded figure. Output of the original data to be carried out with the corresponding explanatory text and formatted output. Individual tasks are shown in Fig. 1.11.

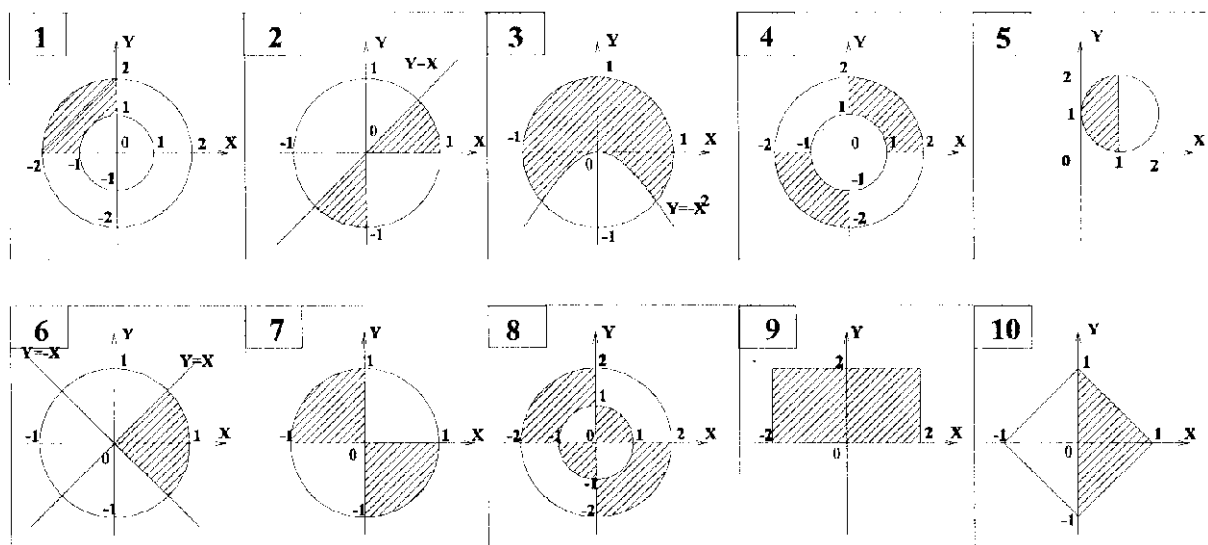


Fig. 1.11 – Individual tasks